



10631 - Aurora's Testimony of Extremely Metal-Poor Galaxies at Cosmic Dawn

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Test	NIRSpec MultiObject Spectroscopy	(2) MSA_EMPG

ABSTRACT

The year 2025 has seen an explosion of discoveries of extremely metal-poor galaxies (EMPGs; $Z/Z_{\text{sun}} < 2\%$) at high redshift, marking a major advance in our pursuit of pristine systems and early chemical enrichment with JWST. However, all metallicity estimates for these EMPGs rely on extrapolations of the R3 calibration, which has never been empirically tested below $Z/Z_{\text{sun}} \sim 2\%$. Whether these relations hold at such low excitation remains entirely unknown—posing a critical uncertainty in interpreting the thousands of hours of JWST observations that have uncovered these sources.

We propose NIRSpec G235H and G395H spectroscopy of the SAHPPIRES EMPG candidates, a flux-complete set of bright, compact galaxies located within a single NIRSpec pointing. This unique configuration allows efficient, high-S/N observations across a representative metallicity range. Our primary goal is to establish a robust R3 calibration in the ultra-low-metallicity regime ($R3 < 2$) using direct measurements of the [O III] 4363 temperature diagnostic (T_e) and [O II] 3726,3729 density indicator (n_e). These observations will define the first reliable abundance scale for EMPGs in the early universe and quantify the deviations from current empirical calibrations.

The resulting calibration will immediately impact all JWST studies of EMPGs by enabling accurate metallicity measurements across the existing high- z samples, and will provide valuable legacy data for other science in the SAP-EDR field, including Little Red Dots and protoclusters out to $z \sim 8.5$.

OBSERVING DESCRIPTION

We request 40 hours of high-resolution NIRSpec MSA spectroscopy using the G235H and G395H gratings (40 hours each) to observe seven EMPG candidates identified in the SAPHIRES-EDR NIRCам pure-parallel survey. The proposed configuration is optimized to obtain robust, temperature-based (“direct”) metallicities and to calibrate strong-line diagnostics at the lowest metallicities observed at high redshift.

MSA Planning Tool simulations show that up to five of the seven EMPG candidates can be observed simultaneously in each configuration. Additional slits will be assigned to serendipitous galaxies in the field, expected to yield 10 $z > 6$ sources and 3 members of the known $z = 8.47$ overdensity, further enhancing the ancillary science return.

This observing strategy maximizes both efficiency and impact by directly coupling pure-parallel EMPG discovery with targeted spectroscopic follow-up. The resulting dataset will provide the first statistical calibration of strong-line diagnostics at $z > 5.5$, establishing a crucial foundation for metallicity studies of early galaxies in the reionization era.

Proposal 10631 - Targets - Aurora's Testimony of Extremely Metal-Poor Galaxies at Cosmic Dawn

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	MSA_EMPG	RA: 04 15 53.3400 (63.9722500d) Dec: -24 09 19.62 (-24.15545d) Equinox: J2000		
	Comments: Description=[]				

Proposal 10631 - Observation 1 - Aurora's Testimony of Extremely Metal-Poor Galaxies at Cosmic Dawn

Observation	Proposal 10631, Observation 1: Test											Fri Mar 13 19:10:41 GMT 2026
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec MultiObject Spectroscopy											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
Acquisition	(2)	MSA_EMPG	RA: 04 15 53.3400 (63.9722500d)									
			Dec: -24 09 19.62 (-24.15545d)									
Template			Equinox: J2000									
	Comments: Description=[]											
Reference Stars	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold				
	MSATA	false	No	MSA Center	MSA_EMPG (161 sources)	MSA_EMPG (161 sources)	jwst-nirspec-hr	1.5				

Proposal 10631 - Observation 1 - Aurora's Testimony of Extremely Metal-Poor Galaxies at Cosmic Dawn

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G235H/F170LP)	c1	3 Shutter Slitlet	63.975282875000 005 Degrees - 24.150852777777 8 Degrees	179.99875696480 53			3	18	25209.602
	2	1 (G235H/F170LP)	c1	3 Shutter Slitlet	63.975282875000 005 Degrees - 24.150852777777 8 Degrees	179.99875696480 53			3	18	25209.602
	3	1 (G235H/F170LP)	c1	3 Shutter Slitlet	63.975282875000 005 Degrees - 24.150852777777 8 Degrees	179.99875696480 53			3	18	25209.602
	4	2 (G395H/F290LP)	c1	3 Shutter Slitlet	63.975282875000 005 Degrees - 24.150852777777 8 Degrees	179.99875696480 53			3	18	25209.602
	5	2 (G395H/F290LP)	c1	3 Shutter Slitlet	63.975282875000 005 Degrees - 24.150852777777 8 Degrees	179.99875696480 53			3	18	25209.602
	6	2 (G395H/F290LP)	c1	3 Shutter Slitlet	63.975282875000 005 Degrees - 24.150852777777 8 Degrees	179.99875696480 53			3	18	25209.602
Special Requirements	Group Visits within 53.0 Days Visits Same PA MSA Planned Aperture PA 180.0000 to 180.0000 Degrees (V3 41.4254303 to 41.4254303)										